# MATH 211 Analysis III — Problem Session 8

## Problem 1

If $z = \sin(x^2 y)$, where $x = st^2$ and $y = s^2 + \frac{1}{t}$, find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$.

**Solution.** By the chain rule,

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s}, \qquad \frac{\partial z}{\partial t} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}.$$

We have

$$z_x = 2xy\cos(x^2y), \qquad z_y = x^2\cos(x^2y),$$

$$x_s = t^2, \qquad x_t = 2st, \qquad y_s = 2s, \qquad y_t = -\frac{1}{t^2}.$$

Thus,

$$\frac{\partial z}{\partial s} = \cos(x^2y)\big(2xy\,t^2 + 2s\,x^2\big), \qquad \frac{\partial z}{\partial t} = \cos(x^2y)\left(4st\,xy - \frac{x^2}{t^2}\right).$$

## Problem 2

Write the appropriate version of the chain rule for $\frac{\partial z}{\partial u}$ if $z = g(x, y)$, where $y = f(x)$ and $x = h(u, v)$.

**Solution.** By the chain rule,

$$\frac{dz}{dx} = \frac{\partial g}{\partial x} + \frac{\partial g}{\partial y}\frac{dy}{dx} = \frac{\partial g}{\partial x} + \frac{\partial g}{\partial y}\,f'(x).$$

Then, since $x = h(u, v)$,

$$\frac{\partial z}{\partial u} = \frac{dz}{dx}\frac{\partial x}{\partial u} = \frac{\partial g}{\partial x}\frac{\partial h}{\partial u} + \frac{\partial g}{\partial y}\,f'(x)\,\frac{\partial h}{\partial u}.$$

## Problem 3

Calculate $\frac{\partial u}{\partial t}$ if $u = \sqrt{x^2 + y^2}$, $x = e^{st}$, $y = 1 + s^2\cos t$.

**Solution.** By the chain rule,

$$\frac{\partial u}{\partial t} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial t},$$

where

$$u_x = \frac{x}{\sqrt{x^2 + y^2}}, \qquad u_y = \frac{y}{\sqrt{x^2 + y^2}}, \qquad x_t = s\,e^{st}, \qquad y_t = -s^2\sin t.$$

Thus,

$$\frac{\partial u}{\partial t} = \frac{x\,s\,e^{st} - y\,s^2\sin t}{\sqrt{x^2 + y^2}} = \frac{s\,e^{2st} - s^2\sin t\,(1 + s^2\cos t)}{\sqrt{e^{2st} + (1 + s^2\cos t)^2}}.$$

**Problem 4**

If $x = t\sin s$ and $y = t\cos s$, find $\frac{\partial^2}{\partial s\,\partial t} f(x,y)$.

**Solution.** We assume $f$ is $C^2$, so $f_{xy} = f_{yx}$. We have

$$x_t = \sin s, \qquad y_t = \cos s, \qquad x_s = t\cos s, \qquad y_s = -t\sin s.$$

First, by the chain rule,

$$\frac{\partial f}{\partial t} = f_x\, x_t + f_y\, y_t = \sin s\, f_x + \cos s\, f_y.$$

Now we differentiate with respect to $s$. Both the coefficients $\sin s$, $\cos s$ and $f_x$, $f_y$ depend on $s$, so we use the product rule, and the chain rule on $f_x$ and $f_y$:

$$\frac{\partial}{\partial s} f_x = f_{xx}\, x_s + f_{xy}\, y_s = t\cos s\, f_{xx} - t\sin s\, f_{xy}, \qquad \frac{\partial}{\partial s} f_y = t\cos s\, f_{yx} - t\sin s\, f_{yy}.$$

Thus,

$$\frac{\partial^2 f}{\partial s\,\partial t} = \cos s\, f_x + \sin s\big(t\cos s\, f_{xx} - t\sin s\, f_{xy}\big) - \sin s\, f_y + \cos s\big(t\cos s\, f_{yx} - t\sin s\, f_{yy}\big).$$

**Problem 5**

If $z = f(x,y)$, where $x = 2s + 3t$ and $y = 3s - 2t$, then find

(a) $\dfrac{\partial^2 z}{\partial s^2}$,

(b) $\dfrac{\partial^2 z}{\partial s\,\partial t}$.

**Solution.** We assume $f$ is $C^2$, so $f_{xy} = f_{yx}$. We have

$$x_s = 2, \qquad y_s = 3, \qquad x_t = 3, \qquad y_t = -2.$$

By the chain rule,

$$\frac{\partial z}{\partial s} = f_x\, x_s + f_y\, y_s = 2f_x + 3f_y.$$

(a) Exercise.

(b) Since $z$ is $C^2$, $\frac{\partial^2 z}{\partial s\,\partial t} = \frac{\partial^2 z}{\partial t\,\partial s}$, so

$$\frac{\partial^2 z}{\partial s\,\partial t} = \frac{\partial}{\partial t}\big(2f_x + 3f_y\big) = 2\big(3f_{xx} - 2f_{xy}\big) + 3\big(3f_{yx} - 2f_{yy}\big) = 6f_{xx} + 5f_{xy} - 6f_{yy}.$$

Observe that the coefficients are constants, so no product rule is needed here, unlike Problem 4.

**Problem 6**

Let $f, g : \mathbb{R} \to \mathbb{R}$ be twice differentiable. Prove that $u(x,y) := f(xy)$ satisfies

$$x\frac{\partial u}{\partial x} - y\frac{\partial u}{\partial y} = 0,$$

and $v(x,y) := f(x-y) + g(x+y)$ satisfies the *wave equation*, i.e.,

$$\frac{\partial^2 v}{\partial x^2} - \frac{\partial^2 v}{\partial y^2} = 0.$$

**Solution.** We first see why $f'$ appears in $u_x$, using the definition of the partial derivative. Note that $f$ is a function of one variable, and in $f(xy)$ its input is the single number $xy$. When $x$ changes to $x+h$ (with $y$ fixed), the input changes from $xy$ to $(x+h)y = xy+hy$, that is, by $hy$. Let $y \neq 0$. Then

$$u_x = \lim_{h\to 0} \frac{f\big((x+h)y\big) - f(xy)}{h} = \lim_{h\to 0} \frac{f(xy+hy) - f(xy)}{hy} \cdot y.$$

Let $k = hy$. Since $y \neq 0$ is fixed, $k \to 0$ as $h \to 0$ and $k \neq 0$ for $h \neq 0$. Thus,

$$u_x = y \cdot \lim_{k\to 0} \frac{f(xy+k) - f(xy)}{k} = y\, f'(xy),$$

where the last limit is exactly the definition of $f'$ at the point $xy$.
So by the argument above $u_x = y\, f'(xy)$. Similarly, $u_y = x\, f'(xy)$. Thus,

$$x\, u_x - y\, u_y = xy\, f'(xy) - xy\, f'(xy) = 0.$$

For $v$, again by the chain rule,

$$v_x = f'(x-y) + g'(x+y), \qquad v_{xx} = f''(x-y) + g''(x+y),$$

$$v_y = -f'(x-y) + g'(x+y), \qquad v_{yy} = f''(x-y) + g''(x+y).$$

Thus,

$$v_{xx} - v_{yy} = \big(f''(x-y) + g''(x+y)\big) - \big(f''(x-y) + g''(x+y)\big) = 0.$$

## Problem 7

Let $f, g : \mathbb{R}^2 \to \mathbb{R}$ be differentiable and satisfy the *Cauchy–Riemann equations*, i.e., that

$$\frac{\partial f}{\partial x} = \frac{\partial g}{\partial y} \qquad \text{and} \qquad \frac{\partial f}{\partial y} = -\frac{\partial g}{\partial x}.$$

If $u(r,\theta) = f(r\cos\theta, r\sin\theta)$ and $v(r,\theta) = g(r\cos\theta, r\sin\theta)$, prove that

$$\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta}, \qquad \frac{\partial v}{\partial r} = -\frac{1}{r}\frac{\partial u}{\partial \theta}, \qquad r \neq 0.$$

**Solution.** Let $x = r\cos\theta$, $y = r\sin\theta$. Then

$$x_r = \cos\theta, \qquad y_r = \sin\theta, \qquad x_\theta = -r\sin\theta, \qquad y_\theta = r\cos\theta.$$

By the chain rule (all partial derivatives of $f$ and $g$ are evaluated at $(r\cos\theta, r\sin\theta)$),

$$u_r = f_x\cos\theta + f_y\sin\theta, \qquad u_\theta = -r\sin\theta\, f_x + r\cos\theta\, f_y,$$

$$v_r = g_x\cos\theta + g_y\sin\theta, \qquad v_\theta = -r\sin\theta\, g_x + r\cos\theta\, g_y.$$

Now we use the Cauchy–Riemann equations $g_y = f_x$ and $g_x = -f_y$. For $r \neq 0$,

$$\frac{1}{r}\, v_\theta = -\sin\theta\, g_x + \cos\theta\, g_y = \sin\theta\, f_y + \cos\theta\, f_x = u_r,$$

$$v_r = g_x\cos\theta + g_y\sin\theta = -f_y\cos\theta + f_x\sin\theta = -\frac{1}{r}\big(-r\sin\theta\, f_x + r\cos\theta\, f_y\big) = -\frac{1}{r}\, u_\theta.$$

## Problem 8

Find the values of the constants $c$ and $k$ for which the function

$$u(x,y) = \frac{1}{1+e^{x-cy}}$$

satisfies the equation

$$u\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = k\frac{\partial^2 u}{\partial x^2}$$

for all $(x,y)$.

**Solution.** Write $u = \big(1+e^{x-cy}\big)^{-1}$. Since $\frac{\partial}{\partial x}e^{x-cy} = e^{x-cy}$ and $\frac{\partial}{\partial y}e^{x-cy} = -c\,e^{x-cy}$, by the chain rule

$$u_x = -\frac{e^{x-cy}}{\big(1+e^{x-cy}\big)^2}, \qquad u_y = \frac{c\,e^{x-cy}}{\big(1+e^{x-cy}\big)^2}.$$

Differentiating $u_x = -e^{x-cy}\big(1+e^{x-cy}\big)^{-2}$ once more with respect to $x$ by the product rule,

$$u_{xx} = -\frac{e^{x-cy}}{\big(1+e^{x-cy}\big)^2} + \frac{2e^{2(x-cy)}}{\big(1+e^{x-cy}\big)^3} = \frac{-e^{x-cy}\big(1+e^{x-cy}\big)+2e^{2(x-cy)}}{\big(1+e^{x-cy}\big)^3} = \frac{e^{2(x-cy)}-e^{x-cy}}{\big(1+e^{x-cy}\big)^3}.$$

Now we compute both sides of the equation. Since $u = \frac{1}{1+e^{x-cy}}$,

$$u\,u_x + u_y = -\frac{e^{x-cy}}{\big(1+e^{x-cy}\big)^3} + \frac{c\,e^{x-cy}\big(1+e^{x-cy}\big)}{\big(1+e^{x-cy}\big)^3} = \frac{c\,e^{2(x-cy)} + (c-1)\,e^{x-cy}}{\big(1+e^{x-cy}\big)^3},$$

$$k\,u_{xx} = \frac{k\,e^{2(x-cy)} - k\,e^{x-cy}}{\big(1+e^{x-cy}\big)^3}.$$

Both sides have the same denominator, so the equation holds if and only if the numerators are equal. Dividing the numerators by $e^{x-cy} \neq 0$, we need

$$c\,e^{x-cy} + c - 1 = k\,e^{x-cy} - k \qquad \text{for all } (x,y).$$

At $(0,0)$ we have $e^{x-cy} = 1$, so $2c-1=0$, hence $c = \frac{1}{2}$. At $(\ln 2, 0)$ we have $e^{x-cy} = 2$, so $3c - 1 = k$, hence $k = \frac{1}{2}$. Conversely, for $c = k = \frac{1}{2}$ both sides are equal to $\frac{1}{2}e^{x-cy} - \frac{1}{2}$ for all $(x,y)$. Thus, $c = \frac{1}{2}$ and $k = \frac{1}{2}$.

Note on the choice of points. Since $c$ is unknown, we take $y = 0$, so that $e^{x-cy} = e^x$ does not depend on $c$. Then we choose $x$ so that $e^x$ is a simple number: $x = 0$ gives $e^0 = 1$ and $x = \ln 2$ gives $e^{\ln 2} = 2$. Any two points with different values of $e^{x-cy}$ would work, since they give two different equations for the two unknowns $c$ and $k$.